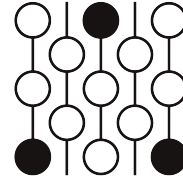


# Review of Submission 5B

First Proof Second Batch

Reviewer 1

June 4–8, 2026



## 1 Review

**Correctness.** The AI has proved the existence of an invariant measure for the SPDE. It does it by studying the invariant measures of the approximate equations and passing to the limit. However, no proof is provided that the invariant measure is unique. The AI states and proves several seemingly standard results from stochastic analysis. Using these results, it formulates a list of conditions that imply uniqueness of the invariant measure. However, it does not prove that any of these conditions hold for the given SPDE.

**Novelty.** The provided solution does not contain any novelty. The existence argument follows a standard approach, and the uniqueness result—which would require additional ideas—has not been proved.

## 2 Recommendation

The solution should be rejected because it is incomplete and does not contain any novel contributions.

## 3 Submission with inline comments

### Overview / Strategy

The original target is to decide whether the Markov semigroup  $P_t$  of the Dirichlet stochastic heat equation with distributional drift  $\delta_0$  has at most one invariant probability measure. The first part of the argument constructs the natural Gibbs measure

$$\nu(dv) = Z^{-1} \exp\left(2 \int_0^1 \mathbf{1}_{\{v(x)>0\}} dx\right) \gamma(dv),$$

where  $\gamma$  is Brownian-bridge law, and proves that  $\nu$  is invariant by passing to the limit from the smooth mollified equations. This is carried out in Theorem 3.3.

Uniqueness requires more than this invariant-measure construction. The clean abstract mechanism is that pairwise non-singularity of one transition skeleton implies uniqueness; see Theorem

3.4. A compact-local Hellinger overlap estimate for the mollified last-step kernels would imply such non-singularity in the limit; see Theorem 3.9. A second route is to prove that every invariant measure is absolutely continuous with respect to  $\nu$ ; see Theorem 3.10 and Proposition 3.12. The stated ABLM-type assumptions provide existence, uniqueness in law, and weak convergence of the mollified equations, but they do not themselves provide any of these deterministic-start smoothing or overlap estimates. Thus the unconditional conclusion proved here is invariance of  $\nu$ , together with precise conditional criteria under which the desired uniqueness follows.

### 3.1 Standing assumptions, notation, and the target statement

Let

$$E := C_0([0, 1]) = \{v \in C([0, 1], \mathbb{R}) : v(0) = v(1) = 0\}, \quad H := L^2(0, 1),$$

with  $E$  equipped with the supremum norm. Let

$$e_k(x) := \sqrt{2} \sin(k\pi x), \quad \lambda_k := \pi^2 k^2, \quad k \geq 1,$$

be the Dirichlet eigenbasis of  $-\partial_{xx}$ . Let  $\gamma$  denote Brownian-bridge law on  $E$ , equivalently the centered Gaussian measure with covariance  $(-\partial_{xx})^{-1}$ :

$$\mathbb{E}_\gamma[(v, e_k)_H (v, e_\ell)_H] = \frac{\delta_{k\ell}}{\lambda_k}.$$

The SPDE from the original problem is

$$\begin{aligned} \frac{\partial u}{\partial t}(x, t) &= \frac{1}{2} \frac{\partial^2 u}{\partial x^2}(x, t) + \delta_0(u(x, t)) + \dot{W}(x, t), \quad t \geq 0, \\ u(x, 0) &= u_0(x) \in E, \\ u(0, t) &= u(1, t) = 0. \end{aligned} \tag{*}$$

**Assumption 3.1** (ABLM-type input on the bounded interval). For every  $u_0 \in E$  and  $T > 0$ , there is a unique-in-law adapted mild solution  $u \in C([0, T], E)$  to (\*), and the solutions define a Markov semigroup  $(P_t)_{t \geq 0}$  on  $E$ . For

$$b_n(r) := G_{1/n}(r) = \sqrt{\frac{n}{2\pi}} e^{-nr^2/2},$$

let  $P_t^n$  be the semigroup of the mollified equation obtained by replacing  $\delta_0$  with  $b_n$ . For every  $t > 0$ ,  $v \in E$ , and  $f \in C_b(E)$ ,

$$P_t^n f(v) \longrightarrow P_t f(v),$$

or equivalently  $P_t^n(v, \cdot) \Rightarrow P_t(v, \cdot)$ .

Define

$$F_n(r) := \int_{-\infty}^r b_n(s) ds, \quad \Phi_n(v) := \int_0^1 F_n(v(x)) dx, \quad \Phi(v) := \int_0^1 \mathbf{1}_{\{v(x) > 0\}} dx.$$

Set

$$\nu_n(dv) := Z_n^{-1} e^{2\Phi_n(v)} \gamma(dv), \quad \nu(dv) := Z^{-1} e^{2\Phi(v)} \gamma(dv), \quad (3.1)$$

where  $Z_n$  and  $Z$  are the normalizing constants. Since  $0 \leq \Phi_n, \Phi \leq 1$ ,

$$1 \leq Z_n, Z \leq e^2, \quad e^{-2} \leq \frac{d\nu_n}{d\gamma}, \frac{d\nu}{d\gamma} \leq e^2. \quad (3.2)$$

For a Markov kernel  $K$  and a probability measure  $\mu$ , write  $\mu K(A) := \int_E K(x, A) \mu(dx)$ . A probability measure  $\mu$  is invariant for a semigroup  $(P_t)$  if  $\mu P_t = \mu$  for all  $t \geq 0$ .

**Theorem 3.2** (Original problem: target and established claim). *The target assertion from the original problem is*

$$(P_t) \text{ has at most one invariant probability measure on } E. \quad (\text{U})$$

*Under Assumption 3.1, the Gibbs measure  $\nu$  in (3.1) is invariant. Moreover, (U) follows if one adds any of the deterministic-start smoothing or overlap hypotheses in Theorem 3.9, Theorem 3.10, or Proposition 3.12 together with Proposition 3.13. In that case the unique invariant probability measure is  $\nu$ .*

*Proof.* The invariance statement is Theorem 3.3. The Hellinger overlap hypothesis of Theorem 3.9 implies pairwise non-singularity of one skeleton, and Theorem 3.4 then gives uniqueness. The compact-local uniform absolute-continuity hypothesis of Theorem 3.10 gives  $\mu \ll \nu$  for every invariant  $\mu$ , and its third hypothesis, supplied for instance by Proposition 3.13, then gives  $\mu = \nu$ . Finally, Proposition 3.12 is a resolvent version of the same absolute-continuity route. Since Theorem 3.3 supplies an invariant measure, any “at most one” conclusion identifies the invariant law with  $\nu$ .  $\square$

### 3.2 The Gibbs invariant measure

**Theorem 3.3** (Invariance of the Gibbs measure). *Under Assumption 3.1, the measure*

$$\nu(dv) = Z^{-1} \exp\left(2 \int_0^1 \mathbf{1}_{\{v(x) > 0\}} dx\right) \gamma(dv)$$

*is invariant for  $(P_t)$ .*

*Proof.* We prove this by approximation.

*Step 1:  $\nu_n \rightarrow \nu$  in total variation.* For each  $r \neq 0$ ,  $F_n(r) \rightarrow \mathbf{1}_{\{r > 0\}}$ . Let  $B \sim \gamma$ . For  $x \in (0, 1)$ ,  $B(x)$  is a nondegenerate centered Gaussian, hence  $\gamma(B(x) = 0) = 0$ . Therefore, by Fubini,

$$\mathbb{E}_\gamma \left| \{x \in [0, 1] : B(x) = 0\} \right| = \int_0^1 \gamma(B(x) = 0) dx = 0,$$

where the endpoints do not matter. Thus  $|\{x : B(x) = 0\}| = 0$  for  $\gamma$ -a.e.  $B$ . For such  $B$ ,

$$\Phi_n(B) = \int_0^1 F_n(B(x)) dx \longrightarrow \int_0^1 \mathbf{1}_{\{B(x) > 0\}} dx = \Phi(B)$$

by dominated convergence in  $x$ . Since  $0 \leq \Phi_n, \Phi \leq 1$ , dominated convergence in  $\gamma$  gives  $Z_n \rightarrow Z$  and

$$\int_E \left| Z_n^{-1} e^{2\Phi_n(v)} - Z^{-1} e^{2\Phi(v)} \right| \gamma(dv) \rightarrow 0.$$

Hence  $\|\nu_n - \nu\|_{\text{TV}} \rightarrow 0$ .

*Step 2:  $\nu_n$  is invariant for  $P_t^n$ .* For fixed  $n$ , the drift  $b_n$  is bounded and Lipschitz. Also  $\Phi_n$  has  $H$ -gradient  $D\Phi_n(v) = b_n(v)$ , because for  $h \in H$ ,

$$D\Phi_n(v)[h] = \int_0^1 b_n(v(x)) h(x) dx = \langle b_n(v), h \rangle_H.$$

On smooth cylinder functions the formal generator is

$$L_n F(v) = \frac{1}{2} \text{Tr } D^2 F(v) - \frac{1}{2} \langle -\partial_{xx} v, DF(v) \rangle_H + \langle D\Phi_n(v), DF(v) \rangle_H.$$

The Gaussian integration-by-parts formula for  $\gamma$ , together with  $d\nu_n = Z_n^{-1} e^{2\Phi_n} d\gamma$ , gives for smooth cylinder  $F, G$

$$\int_E G L_n F d\nu_n = -\frac{1}{2} \int_E \langle DG, DF \rangle_H d\nu_n.$$

Indeed, the derivative of  $e^{2\Phi_n}$  contributes  $2e^{2\Phi_n} D\Phi_n$ , which cancels exactly the added drift term  $\langle D\Phi_n, DF \rangle_H$ . Taking  $G \equiv 1$  gives

$$\int_E L_n F d\nu_n = 0.$$

For bounded Lipschitz gradient drifts, the standard martingale-problem or symmetric-Dirichlet-form construction of the stochastic heat equation identifies  $P_t^n$  with the Markov semigroup generated by the closure of  $L_n$ ; see, for example, [3, 4]. Infinitesimal invariance on the cylinder core therefore implies

$$\nu_n P_t^n = \nu_n, \quad t \geq 0.$$

*Step 3: pass to the singular limit.* Let  $f \in C_b(E)$  and  $t > 0$ . By Assumption 3.1,  $P_t^n f(v) \rightarrow P_t f(v)$  for every  $v \in E$ . Using  $d\nu_n/d\gamma \leq e^2$ , dominated convergence gives

$$\int_E |P_t^n f - P_t f| d\nu_n \leq e^2 \int_E |P_t^n f - P_t f| d\gamma \rightarrow 0.$$

Also,  $\nu_n \rightarrow \nu$  in total variation and  $P_t f$  is bounded, so

$$\int_E P_t f d\nu_n \rightarrow \int_E P_t f d\nu, \quad \int_E f d\nu_n \rightarrow \int_E f d\nu.$$

Since  $\nu_n P_t^n = \nu_n$ ,

$$\int_E P_t f d\nu = \lim_{n \rightarrow \infty} \int_E P_t^n f d\nu_n = \lim_{n \rightarrow \infty} \int_E f d\nu_n = \int_E f d\nu.$$

Thus  $\nu P_t = \nu$  for all  $t > 0$ , and the case  $t = 0$  is trivial.  $\square$

### 3.3 Pairwise overlap and Hellinger criteria

**Theorem 3.4** (Pairwise non-singularity of one skeleton implies uniqueness). *Let  $E$  be Polish and  $K$  a Markov kernel on  $E$ . If*

$$K(x, \cdot) \not\perp K(y, \cdot) \quad \forall x, y \in E,$$

*then  $K$  has at most one invariant probability measure. Consequently, if a semigroup  $(P_t)$  has some  $t_0 > 0$  such that  $P_{t_0}(x, \cdot) \not\perp P_{t_0}(y, \cdot)$  for all  $x, y$ , then  $(P_t)$  has at most one invariant probability measure.*

*Proof.* Assume that  $\mu_1 \neq \mu_2$  are invariant for  $K$ . Let  $\sigma := \mu_1 - \mu_2$ , and write its Jordan decomposition  $\sigma = \sigma^+ - \sigma^-$ . Since  $\sigma(E) = 0$ ,  $m := \sigma^+(E) = \sigma^-(E) > 0$ .

For every finite signed measure  $\eta$ ,

$$|\eta K| \leq |\eta|K.$$

Indeed, for Borel  $A$ , taking the supremum over finite measurable partitions  $\{A_i\}$  of  $A$ ,

$$|\eta K|(A) \leq \sup_{\{A_i\}} \sum_i \int_E K(x, A_i) |\eta|(dx) = \int_E K(x, A) |\eta|(dx).$$

Since  $\sigma K = \sigma$ , we get  $|\sigma| \leq |\sigma|K$ . The two measures have equal total mass, so  $|\sigma|K = |\sigma|$ . Hence

$$\sigma^+ K = \frac{1}{2}(|\sigma|K + \sigma K) = \sigma^+, \quad \sigma^- K = \frac{1}{2}(|\sigma|K - \sigma K) = \sigma^-.$$

Thus  $\alpha := \sigma^+/m$  and  $\beta := \sigma^-/m$  are invariant probabilities, and  $\alpha \perp \beta$ . Choose a Borel set  $A$  with  $\alpha(A) = 1$  and  $\beta(A) = 0$ . Invariance gives

$$1 = \int_E K(x, A) \alpha(dx), \quad 0 = \int_E K(y, A) \beta(dy).$$

Therefore  $K(x, A) = 1$  for  $\alpha$ -a.e.  $x$  and  $K(y, A) = 0$  for  $\beta$ -a.e.  $y$ . Choosing such  $x, y$ ,  $K(x, \cdot)$  is concentrated on  $A$  while  $K(y, \cdot)$  is concentrated on  $A^c$ , so  $K(x, \cdot) \perp K(y, \cdot)$ , a contradiction. The semigroup assertion follows because every invariant measure for  $(P_t)$  is invariant for the skeleton  $P_{t_0}$ .  $\square$

**Definition 3.5** (Hellinger affinity). For probability measures  $\mu, \rho$  on  $E$ , define

$$A(\mu, \rho) := \int_E \sqrt{\frac{d\mu}{d\lambda} \frac{d\rho}{d\lambda}} d\lambda,$$

where  $\lambda$  is any finite measure dominating both  $\mu$  and  $\rho$ . The value is independent of  $\lambda$ . We use the normalized total variation norm

$$\|\mu - \rho\|_{\text{TV}} := \sup_{A \in \mathcal{B}(E)} |\mu(A) - \rho(A)|.$$

**Lemma 3.6** (Elementary Hellinger facts). *For probability measures  $\mu, \rho$ ,*

$$\mu \perp \rho \iff \|\mu - \rho\|_{\text{TV}} = 1 \iff A(\mu, \rho) = 0,$$

and

$$\|\mu - \rho\|_{\text{TV}} \leq \sqrt{1 - A(\mu, \rho)^2}.$$

Moreover, if  $\mathcal{P}$  ranges over finite measurable partitions of  $E$ , then

$$A(\mu, \rho) = \inf_{\mathcal{P}} \sum_{B \in \mathcal{P}} \sqrt{\mu(B)\rho(B)}.$$

*Proof.* Take  $\lambda = \mu + \rho$ , and write  $f = d\mu/d\lambda$ ,  $g = d\rho/d\lambda$ . The definition is independent of  $\lambda$  because changing the dominating measure multiplies both densities by the same Radon–Nikodym factor. If  $\mu \perp \rho$ , then  $fg = 0$   $\lambda$ -a.e., so  $A(\mu, \rho) = 0$ . Conversely, if  $A(\mu, \rho) = 0$ , then  $\sqrt{fg} = 0$   $\lambda$ -a.e.; the sets  $\{f > 0\}$  and  $\{g > 0\}$  separate  $\mu$  and  $\rho$ , so  $\mu \perp \rho$ . The equivalence with  $\|\mu - \rho\|_{\text{TV}} = 1$  is the standard characterization of singularity for probability measures.

Also,

$$\begin{aligned} \|\mu - \rho\|_{\text{TV}} &= \frac{1}{2} \int_E |f - g| d\lambda \\ &= \frac{1}{2} \int_E |\sqrt{f} - \sqrt{g}|(\sqrt{f} + \sqrt{g}) d\lambda \\ &\leq \frac{1}{2} \left( \int_E (\sqrt{f} - \sqrt{g})^2 d\lambda \right)^{1/2} \left( \int_E (\sqrt{f} + \sqrt{g})^2 d\lambda \right)^{1/2} \\ &= \sqrt{1 - A(\mu, \rho)^2}. \end{aligned}$$

For the partition formula, let  $\mathcal{P}$  be finite. On each atom  $B$ ,

$$\int_B \sqrt{\mathbb{E}_\lambda(f | \mathcal{P}) \mathbb{E}_\lambda(g | \mathcal{P})} d\lambda = \sqrt{\mu(B)\rho(B)}.$$

Conditional Cauchy–Schwarz gives

$$\mathbb{E}_\lambda(\sqrt{fg} | \mathcal{P}) \leq \sqrt{\mathbb{E}_\lambda(f | \mathcal{P}) \mathbb{E}_\lambda(g | \mathcal{P})},$$

so every partition sum is at least  $A(\mu, \rho)$ . Since  $E$  is Polish,  $\mathcal{B}(E)$  is countably generated. Choosing finite partitions increasing to a countable generating algebra, the bounded martingale convergence theorem (applied with  $\lambda = \mu + \rho$ , so  $0 \leq f, g \leq 1$ ) gives partition sums converging down to  $\int \sqrt{fg} d\lambda$ . Hence the infimum equals  $A(\mu, \rho)$ .  $\square$

**Lemma 3.7** (Mixture lower bound). *Let  $K$  be a Markov kernel on  $E$ . Suppose  $C \subset E$  is measurable and there is  $a > 0$  such that*

$$A(K(v, \cdot), K(w, \cdot)) \geq a \quad \forall v, w \in C.$$

Then, for all probability measures  $\alpha, \beta$ ,

$$A(\alpha K, \beta K) \geq a \alpha(C) \beta(C).$$

*Proof.* Let  $\mathcal{P}$  be a finite partition. For each atom  $B$ ,

$$\sqrt{(\alpha K)(B)(\beta K)(B)} \geq \int_C \int_C \sqrt{K(v, B)K(w, B)} \alpha(dv) \beta(dw),$$

by Cauchy–Schwarz applied to the functions  $\sqrt{K(v, B)}$  and  $\sqrt{K(w, B)}$ . Summing over  $B \in \mathcal{P}$  gives

$$\sum_{B \in \mathcal{P}} \sqrt{(\alpha K)(B)(\beta K)(B)} \geq \int_C \int_C \sum_{B \in \mathcal{P}} \sqrt{K(v, B)K(w, B)} \alpha(dv) \beta(dw).$$

The inner sum is at least  $A(K(v, \cdot), K(w, \cdot))$  by Lemma 3.6; hence the last display is at least  $a\alpha(C)\beta(C)$ . Taking the infimum over  $\mathcal{P}$  and using the partition formula proves the claim.  $\square$

**Lemma 3.8** (Lower semicontinuity of total variation). *If  $\mu_n \Rightarrow \mu$  and  $\rho_n \Rightarrow \rho$  weakly on a Polish space, then*

$$\|\mu - \rho\|_{\text{TV}} \leq \liminf_{n \rightarrow \infty} \|\mu_n - \rho_n\|_{\text{TV}}.$$

*Proof.* For  $\eta = \mu - \rho$ , regularity of the finite signed measure  $\eta$  implies

$$\|\eta\|_{\text{TV}} = \sup_{\substack{f \in C_b(E) \\ 0 \leq f \leq 1}} \left| \int_E f d\eta \right|.$$

Indeed, indicators of Borel sets are approximated from inside and outside by continuous functions using regularity of  $|\eta|$  and Urysohn’s lemma. For each such  $f$ ,

$$\int_E f d(\mu_n - \rho_n) \longrightarrow \int_E f d(\mu - \rho),$$

so

$$\left| \int_E f d(\mu - \rho) \right| \leq \liminf_n \|\mu_n - \rho_n\|_{\text{TV}}.$$

Taking the supremum over  $f$  proves the result.  $\square$

**Theorem 3.9** (Conditional uniqueness from compact-local Hellinger overlap). *Let  $E$  be Polish. Let  $(P_s)$  and  $(P_s^n)$  be Markov semigroups on  $E$ . Fix  $0 < \varepsilon < t$ . Assume:*

1. *for every  $x \in E$ ,*

$$P_t^n(x, \cdot) \Rightarrow P_t(x, \cdot), \quad P_{t-\varepsilon}^n(x, \cdot) \Rightarrow P_{t-\varepsilon}(x, \cdot);$$

2. *for every compact  $C \subset E$ , there is  $a(C) > 0$  such that*

$$A(P_\varepsilon^n(v, \cdot), P_\varepsilon^n(w, \cdot)) \geq a(C)$$

*for all  $n \geq 1$  and all  $v, w \in C$ .*

*Then  $P_t(x, \cdot) \not\equiv P_t(y, \cdot)$  for all  $x, y \in E$ . Consequently, if this holds for some  $t > 0$ , then  $(P_s)$  has at most one invariant probability measure.*

*Proof.* Fix  $x, y \in E$ , and set

$$\alpha_n := P_{t-\varepsilon}^n(x, \cdot), \quad \beta_n := P_{t-\varepsilon}^n(y, \cdot).$$

The two sequences are tight, because they are weakly convergent. Hence there is a compact  $C \subset E$  such that  $\alpha_n(C) \geq 1/2$  and  $\beta_n(C) \geq 1/2$  for all  $n$ . By hypothesis 2,

$$A(P_\varepsilon^n(v, \cdot), P_\varepsilon^n(w, \cdot)) \geq a(C), \quad v, w \in C.$$

Using the semigroup property and Lemma 3.7,

$$A(P_t^n(x, \cdot), P_t^n(y, \cdot)) = A(\alpha_n P_\varepsilon^n, \beta_n P_\varepsilon^n) \geq a(C) \alpha_n(C) \beta_n(C) \geq \frac{a(C)}{4}.$$

By Lemma 3.6,

$$\|P_t^n(x, \cdot) - P_t^n(y, \cdot)\|_{\text{TV}} \leq \sqrt{1 - \left(\frac{a(C)}{4}\right)^2} =: \rho < 1.$$

Weak convergence at time  $t$ , followed by Lemma 3.8, yields

$$\|P_t(x, \cdot) - P_t(y, \cdot)\|_{\text{TV}} \leq \rho < 1.$$

Thus the two limiting transition probabilities are not singular. The final claim follows from Theorem 3.4.  $\square$

### 3.4 Absolute continuity routes and the Dirichlet-form limitation

**Theorem 3.10** (Conditional uniqueness from compact-local uniform absolute continuity). *Let  $E = C_0([0, 1])$ , and let  $\nu$  be the Gibbs measure in (3.1). Fix  $t_0 > 0$ . Assume:*

1. *for every  $x \in E$  and  $f \in C_b(E)$ ,*

$$P_{t_0}^n f(x) \rightarrow P_{t_0} f(x);$$

2. *for every compact  $K \subset E$ , the family*

$$\{P_{t_0}^n(x, \cdot) : n \geq 1, x \in K\}$$

*is uniformly absolutely continuous with respect to  $\nu$ ;*

3. *every  $P_t$ -invariant probability measure that is absolutely continuous with respect to  $\nu$  equals  $\nu$ .*

*Then  $\nu$  is the unique invariant probability measure of  $(P_t)$ .*



*Proof.* Let  $\mu$  be invariant for  $(P_t)$ , and set  $\rho_n := \mu P_{t_0}^n$ . For  $f \in C_b(E)$ , dominated convergence and hypothesis 1 give

$$\int_E f d\rho_n = \int_E P_{t_0}^n f(x) \mu(dx) \longrightarrow \int_E P_{t_0} f(x) \mu(dx) = \int_E f d\mu,$$

so  $\rho_n \Rightarrow \mu$ .

Let  $B$  be Borel with  $\nu(B) = 0$ . Fix  $\varepsilon, \eta > 0$ . By tightness of  $\mu$ , choose compact  $K$  with  $\mu(K^c) < \eta$ . By uniform absolute continuity on  $K$ , choose  $\delta > 0$  such that

$$\nu(A) < \delta \implies \sup_{n \geq 1} \sup_{x \in K} P_{t_0}^n(x, A) < \varepsilon.$$

By outer regularity, choose open  $O \supset B$  with  $\nu(O) < \delta$ . Then for every  $n$ ,

$$\rho_n(O) = \int_E P_{t_0}^n(x, O) \mu(dx) \leq \varepsilon + \eta.$$

Since  $O$  is open and  $\rho_n \Rightarrow \mu$ , Portmanteau gives

$$\mu(O) \leq \liminf_n \rho_n(O) \leq \varepsilon + \eta.$$

Thus  $\mu(B) \leq \varepsilon + \eta$ . Letting  $\varepsilon, \eta \downarrow 0$  shows  $\mu \ll \nu$ . Hypothesis 3 gives  $\mu = \nu$ .  $\square$

*Remark 3.11* (A simple sufficient condition for uniform absolute continuity). Hypothesis 2 of Theorem 3.10 follows if, for every compact  $K \subset E$ , there are  $p > 1$  and  $C_K < \infty$  such that  $P_{t_0}^n(x, \cdot) = r_{n,x}\nu$  and

$$\sup_{n \geq 1} \sup_{x \in K} \|r_{n,x}\|_{L^p(\nu)} \leq C_K.$$

Indeed, with  $q = p/(p-1)$ , Holder's inequality gives  $P_{t_0}^n(x, A) \leq C_K \nu(A)^{1/q}$ .

**Proposition 3.12** (Pointwise resolvent absolute continuity). *Assume that  $\nu$  is invariant and that every invariant probability measure absolutely continuous with respect to  $\nu$  equals  $\nu$ . If, for some  $\lambda > 0$ ,*

$$R_\lambda(x, \cdot) := \int_0^\infty e^{-\lambda t} P_t(x, \cdot) dt \ll \nu \quad \forall x \in E,$$

*then  $\nu$  is the unique invariant probability measure.*

*Proof.* Let  $\mu$  be invariant and let  $B$  be Borel with  $\nu(B) = 0$ . Then  $R_\lambda(x, B) = 0$  for every  $x$ , and therefore

$$0 = \int_E R_\lambda(x, B) \mu(dx) = \int_0^\infty e^{-\lambda t} \mu P_t(B) dt = \int_0^\infty e^{-\lambda t} \mu(B) dt = \frac{1}{\lambda} \mu(B).$$

Thus  $\mu \ll \nu$ , and by the assumed uniqueness among  $\nu$ -absolutely continuous invariant measures,  $\mu = \nu$ .  $\square$

**Proposition 3.13** (What the Gibbs Dirichlet form proves). *Let  $(T_t)$  be a  $\nu$ -symmetric Markov semigroup on  $L^2(\nu)$ , and let  $\mathcal{E}$  be its closed Dirichlet form. Assume that the only zero-energy functions in the domain of  $\mathcal{E}$  are constants. Then every  $T_t$ -invariant probability measure  $\mu$  with  $\mu \ll \nu$  equals  $\nu$ .*

*For the Gibbs measure above, the zero-energy condition holds for the gradient form*

$$\mathcal{E}(F, G) = \frac{1}{2} \int_E \langle DF, DG \rangle_H d\nu$$

*provided the limiting semigroup is indeed the symmetric semigroup associated with this form.*

*Proof.* Let  $h = d\mu/d\nu$ . Invariance gives, for bounded  $f$ ,

$$\int_E T_t f h d\nu = \int_E f h d\nu.$$

By symmetry and the  $L^1$ -extension of  $T_t$ , this says  $T_t h = h$  in  $L^1(\nu)$ . For  $m \geq 1$ , let  $h_m = h \wedge m$ . Since  $r \mapsto r \wedge m$  is concave and  $T_t$  is Markov,

$$T_t h_m \leq (T_t h) \wedge m = h_m.$$

Integrating both sides against the invariant measure  $\nu$  gives equality, hence  $T_t h_m = h_m$ . Now  $h_m \in L^2(\nu)$ , and a fixed  $L^2$ -function has zero energy:

$$\mathcal{E}(h_m, h_m) = \lim_{t \downarrow 0} \frac{1}{t} \langle h_m - T_t h_m, h_m \rangle_{L^2(\nu)} = 0.$$

Thus  $h_m$  is constant. By monotone convergence,  $\int h_m d\nu \rightarrow \int h d\nu = 1$ , so  $h_m \rightarrow 1$  and  $h = 1$   $\nu$ -a.e.

For the displayed Gibbs gradient form, the density bounds (3.2) make  $\nu$  equivalent to  $\gamma$  with bounded density above and below. The Gaussian Poincare inequality for the Brownian-bridge measure,

$$\text{Var}_\gamma(F) \leq \frac{1}{\pi^2} \int_E \|DF\|_H^2 d\gamma$$

on cylinder functions, extends by closure. Hence zero  $\nu$ -energy implies zero  $\gamma$ -energy and therefore constancy.  $\square$

**Example 3.14** (Irreducibility in  $L^2(\nu)$  does not rule out singular invariant laws). Let  $E = [0, 1]$ , let  $\nu$  be Lebesgue measure, and define a Markov kernel

$$K(x, \cdot) = \begin{cases} \nu(\cdot), & x \neq 0, \\ \delta_0(\cdot), & x = 0. \end{cases}$$

Then  $\nu K = \nu$  because  $\nu(\{0\}) = 0$ , and  $\delta_0 K = \delta_0$ . On  $L^2(\nu)$ , however,  $Kf = \int f d\nu$   $\nu$ -a.e.; hence the  $L^2(\nu)$  operator is symmetric and has only constant fixed functions. The extra invariant measure  $\delta_0$  is invisible to the  $L^2(\nu)$  theory because it is supported on a  $\nu$ -null point.

### 3.5 What is missing from the stated assumptions

The preceding results identify exactly what would close the uniqueness proof. One needs a deterministic-start smoothing or overlap estimate, such as:

$$P_{t_0}(x, \cdot) \not\perp P_{t_0}(y, \cdot) \quad \forall x, y,$$

or the compact-local Hellinger estimate of Theorem 3.9, or compact-local uniform  $\nu$ -absolute continuity as in Theorem 3.10, or pointwise resolvent absolute continuity as in Proposition 3.12. These are not consequences of weak convergence alone.

**Example 3.15** (Weak convergence can destroy absolute continuity and overlap). On  $\mathbb{R}$ , each Gaussian law  $\mathcal{N}(0, 1/n)$  is absolutely continuous with respect to Lebesgue measure, but

$$\mathcal{N}(0, 1/n) \Rightarrow \delta_0,$$

and  $\delta_0$  is singular with respect to Lebesgue measure. Likewise,  $\mathcal{N}(0, 1/n)$  and  $\mathcal{N}(1, 1/n)$  have positive Hellinger affinity for each  $n$ , but their affinities tend to 0, and the weak limits  $\delta_0$  and  $\delta_1$  are singular. Thus weak convergence of kernels does not by itself preserve absolute continuity, uniform integrability, or a positive overlap lower bound.

*Remark 3.16* (Finite-dimensional domination as a possible missing theorem). A convenient way to prove the compact-local Hellinger input would be a uniform finite-dimensional heat-kernel domination theorem. Let  $P_t^{n,m}$  be the  $m$ -mode Galerkin approximation of the mollified equation and  $Q_t^m$  the  $m$ -dimensional Dirichlet Ornstein–Uhlenbeck kernel. A bound of the form

$$P_t^{n,m}(a, \cdot) \geq c Q_t^m(a, \cdot),$$

with  $c > 0$  independent of  $n, m, a$ , together with a passage from Galerkin kernels to the infinite-dimensional kernels, would yield overlap. For the Galerkin potential

$$\Psi_{n,m}(a) = \int_0^1 F_n \left( \sum_{k=1}^m a_k e_k(x) \right) dx,$$

one has the layer-cake representation, with  $e(x) = (e_1(x), \dots, e_m(x))$ ,

$$\Psi_{n,m}(a) = \int_0^1 \int_{\mathbb{R}} \mathbf{1}_{\{\langle e(x), a \rangle > s\}} b_n(s) ds \, dx.$$

This displays  $\Psi_{n,m}$  as an average of affine half-space membrane potentials of total mass 1. Proving the domination estimate is, however, an additional finite-dimensional heat-kernel theorem; it is not part of Assumption 3.1.

## Final conclusion

The natural Gibbs measure  $\nu$  is rigorously proved invariant for the limiting singular semigroup. The desired uniqueness statement follows from any of the deterministic-start smoothing or overlap estimates isolated above. Under only the assumptions explicitly stated in the original problem—namely well-posedness and weak convergence of the mollified equations—the argument does not rule out invariant measures singular with respect to  $\nu$ . Thus the candidate proof establishes invariance and conditional uniqueness, but an additional smoothing or overlap theorem is needed for an unconditional proof of (U).

## References

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- [4] Z.-M. Ma and M. Röckner, *Introduction to the Theory of (Non-Symmetric) Dirichlet Forms*, Universitext, Springer, 1992.
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The article has been published

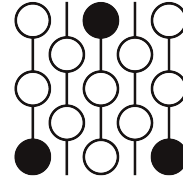
Reviewer 1

# Review of Submission 5B

First Proof Second Batch

Reviewer 2

June 4–8, 2026



## 1 Review

**Correctness.** The solution is explicitly partial. The invariance of the explicit Gibbs measure is proved (but should be cited to the literature, as it was already known). The submission claims to provide a number of results towards uniqueness conditional on various assumptions, which are essentially standard in the literature. None of these parts of the submission engage at all with the specific form of the equation under consideration, which is required for a proof. In other words, these are abstract criteria for the uniqueness of an invariant measure, and no progress is made towards checking the conditions in the specific situation under consideration.

**Novelty.** I do not believe that any of the criteria for uniqueness are novel.

## 2 Recommendation

I recommend that this submission be rejected as it does not make any serious progress towards attacking the main difficulty of the problem.

## 3 Submission with inline comments

### Overview / Strategy

The original target is to decide whether the Markov semigroup  $P_t$  of the Dirichlet stochastic heat equation with distributional drift  $\delta_0$  has at most one invariant probability measure. The first part of the argument constructs the natural Gibbs measure

$$\nu(dv) = Z^{-1} \exp\left(2 \int_0^1 \mathbf{1}_{\{v(x) > 0\}} dx\right) \gamma(dv),$$

where  $\gamma$  is Brownian-bridge law, and proves that  $\nu$  is invariant by passing to the limit from the smooth mollified equations. This is carried out in Theorem 3.3.

Uniqueness requires more than this invariant-measure construction. The clean abstract mechanism is that pairwise non-singularity of one transition skeleton implies uniqueness; see Theorem

3.4. A compact-local Hellinger overlap estimate for the mollified last-step kernels would imply such non-singularity in the limit; see Theorem 3.9. A second route is to prove that every invariant measure is absolutely continuous with respect to  $\nu$ ; see Theorem 3.10 and Proposition 3.12. The stated ABLM-type assumptions provide existence, uniqueness in law, and weak convergence of the mollified equations, but they do not themselves provide any of these deterministic-start smoothing or overlap estimates. Thus the unconditional conclusion proved here is invariance of  $\nu$ , together with precise conditional criteria under which the desired uniqueness follows.

### 3.1 Standing assumptions, notation, and the target statement

Let

$$E := C_0([0, 1]) = \{v \in C([0, 1], \mathbb{R}) : v(0) = v(1) = 0\}, \quad H := L^2(0, 1),$$

with  $E$  equipped with the supremum norm. Let

$$e_k(x) := \sqrt{2} \sin(k\pi x), \quad \lambda_k := \pi^2 k^2, \quad k \geq 1,$$

be the Dirichlet eigenbasis of  $-\partial_{xx}$ . Let  $\gamma$  denote Brownian-bridge law on  $E$ , equivalently the centered Gaussian measure with covariance  $(-\partial_{xx})^{-1}$ :

$$\mathbb{E}_\gamma[(v, e_k)_H (v, e_\ell)_H] = \frac{\delta_{k\ell}}{\lambda_k}.$$

The SPDE from the original problem is

$$\begin{aligned} \frac{\partial u}{\partial t}(x, t) &= \frac{1}{2} \frac{\partial^2 u}{\partial x^2}(x, t) + \delta_0(u(x, t)) + \dot{W}(x, t), \quad t \geq 0, \\ u(x, 0) &= u_0(x) \in E, \\ u(0, t) &= u(1, t) = 0. \end{aligned} \tag{*}$$

**Assumption 3.1** (ABLM-type input on the bounded interval). For every  $u_0 \in E$  and  $T > 0$ , there is a unique-in-law adapted mild solution  $u \in C([0, T], E)$  to (\*), and the solutions define a Markov semigroup  $(P_t)_{t \geq 0}$  on  $E$ . For

$$b_n(r) := G_{1/n}(r) = \sqrt{\frac{n}{2\pi}} e^{-nr^2/2},$$

let  $P_t^n$  be the semigroup of the mollified equation obtained by replacing  $\delta_0$  with  $b_n$ . For every  $t > 0$ ,  $v \in E$ , and  $f \in C_b(E)$ ,

$$P_t^n f(v) \longrightarrow P_t f(v),$$

or equivalently  $P_t^n(v, \cdot) \Rightarrow P_t(v, \cdot)$ .

Define

$$F_n(r) := \int_{-\infty}^r b_n(s) ds, \quad \Phi_n(v) := \int_0^1 F_n(v(x)) dx, \quad \Phi(v) := \int_0^1 \mathbf{1}_{\{v(x) > 0\}} dx.$$

Set

$$\nu_n(dv) := Z_n^{-1} e^{2\Phi_n(v)} \gamma(dv), \quad \nu(dv) := Z^{-1} e^{2\Phi(v)} \gamma(dv), \quad (3.1)$$

where  $Z_n$  and  $Z$  are the normalizing constants. Since  $0 \leq \Phi_n, \Phi \leq 1$ ,

$$1 \leq Z_n, Z \leq e^2, \quad e^{-2} \leq \frac{d\nu_n}{d\gamma}, \frac{d\nu}{d\gamma} \leq e^2. \quad (3.2)$$

For a Markov kernel  $K$  and a probability measure  $\mu$ , write  $\mu K(A) := \int_E K(x, A) \mu(dx)$ . A probability measure  $\mu$  is invariant for a semigroup  $(P_t)$  if  $\mu P_t = \mu$  for all  $t \geq 0$ .

**Theorem 3.2** (Original problem: target and established claim). *The target assertion from the original problem is*

$$(P_t) \text{ has at most one invariant probability measure on } E. \quad (\text{U})$$

*Under Assumption 3.1, the Gibbs measure  $\nu$  in (3.1) is invariant. Moreover, (U) follows if one adds any of the deterministic-start smoothing or overlap hypotheses in Theorem 3.9, Theorem 3.10, or Proposition 3.12 together with Proposition 3.13. In that case the unique invariant probability measure is  $\nu$ .*

*Proof.* The invariance statement is Theorem 3.3. The Hellinger overlap hypothesis of Theorem 3.9 implies pairwise non-singularity of one skeleton, and Theorem 3.4 then gives uniqueness. The compact-local uniform absolute-continuity hypothesis of Theorem 3.10 gives  $\mu \ll \nu$  for every invariant  $\mu$ , and its third hypothesis, supplied for instance by Proposition 3.13, then gives  $\mu = \nu$ . Finally, Proposition 3.12 is a resolvent version of the same absolute-continuity route. Since Theorem 3.3 supplies an invariant measure, any “at most one” conclusion identifies the invariant law with  $\nu$ .  $\square$

## 3.2 The Gibbs invariant measure

**Theorem 3.3** (Invariance of the Gibbs measure). *Under Assumption 3.1, the measure*

$$\nu(dv) = Z^{-1} \exp\left(2 \int_0^1 \mathbf{1}_{\{v(x) > 0\}} dx\right) \gamma(dv)$$

*is invariant for  $(P_t)$ .*

Reviewer 2

This should be cited to Bounebaché–Zambotti (2014).

*Proof.* We prove this by approximation.

*Step 1:  $\nu_n \rightarrow \nu$  in total variation.* For each  $r \neq 0$ ,  $F_n(r) \rightarrow \mathbf{1}_{\{r > 0\}}$ . Let  $B \sim \gamma$ . For  $x \in (0, 1)$ ,  $B(x)$  is a nondegenerate centered Gaussian, hence  $\gamma(B(x) = 0) = 0$ . Therefore, by Fubini,

$$\mathbb{E}_\gamma \left| \{x \in [0, 1] : B(x) = 0\} \right| = \int_0^1 \gamma(B(x) = 0) dx = 0,$$

where the endpoints do not matter. Thus  $|\{x : B(x) = 0\}| = 0$  for  $\gamma$ -a.e.  $B$ . For such  $B$ ,

$$\Phi_n(B) = \int_0^1 F_n(B(x)) dx \longrightarrow \int_0^1 \mathbf{1}_{\{B(x) > 0\}} dx = \Phi(B)$$

by dominated convergence in  $x$ . Since  $0 \leq \Phi_n, \Phi \leq 1$ , dominated convergence in  $\gamma$  gives  $Z_n \rightarrow Z$  and

$$\int_E |Z_n^{-1} e^{2\Phi_n(v)} - Z^{-1} e^{2\Phi(v)}| \gamma(dv) \longrightarrow 0.$$

Hence  $\|\nu_n - \nu\|_{\text{TV}} \rightarrow 0$ .

*Step 2:  $\nu_n$  is invariant for  $P_t^n$ .* For fixed  $n$ , the drift  $b_n$  is bounded and Lipschitz. Also  $\Phi_n$  has  $H$ -gradient  $D\Phi_n(v) = b_n(v)$ , because for  $h \in H$ ,

$$D\Phi_n(v)[h] = \int_0^1 b_n(v(x))h(x) dx = \langle b_n(v), h \rangle_H.$$

On smooth cylinder functions the formal generator is

$$L_n F(v) = \frac{1}{2} \text{Tr} D^2 F(v) - \frac{1}{2} \langle -\partial_{xx} v, DF(v) \rangle_H + \langle D\Phi_n(v), DF(v) \rangle_H.$$

The Gaussian integration-by-parts formula for  $\gamma$ , together with  $d\nu_n = Z_n^{-1} e^{2\Phi_n} d\gamma$ , gives for smooth cylinder  $F, G$

$$\int_E G L_n F d\nu_n = -\frac{1}{2} \int_E \langle DG, DF \rangle_H d\nu_n.$$

Indeed, the derivative of  $e^{2\Phi_n}$  contributes  $2e^{2\Phi_n} D\Phi_n$ , which cancels exactly the added drift term  $\langle D\Phi_n, DF \rangle_H$ . Taking  $G \equiv 1$  gives

$$\int_E L_n F d\nu_n = 0.$$

For bounded Lipschitz gradient drifts, the standard martingale-problem or symmetric-Dirichlet-form construction of the stochastic heat equation identifies  $P_t^n$  with the Markov semigroup generated by the closure of  $L_n$ ; see, for example, [3, 4]. Infinitesimal invariance on the cylinder core therefore implies

$$\nu_n P_t^n = \nu_n, \quad t \geq 0.$$

*Step 3: pass to the singular limit.* Let  $f \in C_b(E)$  and  $t > 0$ . By Assumption 3.1,  $P_t^n f(v) \rightarrow P_t f(v)$  for every  $v \in E$ . Using  $d\nu_n/d\gamma \leq e^2$ , dominated convergence gives

$$\int_E |P_t^n f - P_t f| d\nu_n \leq e^2 \int_E |P_t^n f - P_t f| d\gamma \longrightarrow 0.$$

Also,  $\nu_n \rightarrow \nu$  in total variation and  $P_t f$  is bounded, so

$$\int_E P_t f d\nu_n \longrightarrow \int_E P_t f d\nu, \quad \int_E f d\nu_n \longrightarrow \int_E f d\nu.$$

Since  $\nu_n P_t^n = \nu_n$ ,

$$\int_E P_t f d\nu = \lim_{n \rightarrow \infty} \int_E P_t^n f d\nu_n = \lim_{n \rightarrow \infty} \int_E f d\nu_n = \int_E f d\nu.$$

Thus  $\nu P_t = \nu$  for all  $t > 0$ , and the case  $t = 0$  is trivial.  $\square$



### 3.3 Pairwise overlap and Hellinger criteria

**Theorem 3.4** (Pairwise non-singularity of one skeleton implies uniqueness). *Let  $E$  be Polish and  $K$  a Markov kernel on  $E$ . If*

$$K(x, \cdot) \not\perp K(y, \cdot) \quad \forall x, y \in E,$$

*then  $K$  has at most one invariant probability measure. Consequently, if a semigroup  $(P_t)$  has some  $t_0 > 0$  such that  $P_{t_0}(x, \cdot) \not\perp P_{t_0}(y, \cdot)$  for all  $x, y$ , then  $(P_t)$  has at most one invariant probability measure.*

*Proof.* Assume that  $\mu_1 \neq \mu_2$  are invariant for  $K$ . Let  $\sigma := \mu_1 - \mu_2$ , and write its Jordan decomposition  $\sigma = \sigma^+ - \sigma^-$ . Since  $\sigma(E) = 0$ ,  $m := \sigma^+(E) = \sigma^-(E) > 0$ .

For every finite signed measure  $\eta$ ,

$$|\eta K| \leq |\eta|K.$$

Indeed, for Borel  $A$ , taking the supremum over finite measurable partitions  $\{A_i\}$  of  $A$ ,

$$|\eta K|(A) \leq \sup_{\{A_i\}} \sum_i \int_E K(x, A_i) |\eta|(dx) = \int_E K(x, A) |\eta|(dx).$$

Since  $\sigma K = \sigma$ , we get  $|\sigma| \leq |\sigma|K$ . The two measures have equal total mass, so  $|\sigma|K = |\sigma|$ . Hence

$$\sigma^+ K = \frac{1}{2}(|\sigma|K + \sigma K) = \sigma^+, \quad \sigma^- K = \frac{1}{2}(|\sigma|K - \sigma K) = \sigma^-.$$

Thus  $\alpha := \sigma^+/m$  and  $\beta := \sigma^-/m$  are invariant probabilities, and  $\alpha \perp \beta$ . Choose a Borel set  $A$  with  $\alpha(A) = 1$  and  $\beta(A) = 0$ . Invariance gives

$$1 = \int_E K(x, A) \alpha(dx), \quad 0 = \int_E K(y, A) \beta(dy).$$

Therefore  $K(x, A) = 1$  for  $\alpha$ -a.e.  $x$  and  $K(y, A) = 0$  for  $\beta$ -a.e.  $y$ . Choosing such  $x, y$ ,  $K(x, \cdot)$  is concentrated on  $A$  while  $K(y, \cdot)$  is concentrated on  $A^c$ , so  $K(x, \cdot) \perp K(y, \cdot)$ , a contradiction. The semigroup assertion follows because every invariant measure for  $(P_t)$  is invariant for the skeleton  $P_{t_0}$ .  $\square$

**Definition 3.5** (Hellinger affinity). For probability measures  $\mu, \rho$  on  $E$ , define

$$A(\mu, \rho) := \int_E \sqrt{\frac{d\mu}{d\lambda} \frac{d\rho}{d\lambda}} d\lambda,$$

where  $\lambda$  is any finite measure dominating both  $\mu$  and  $\rho$ . The value is independent of  $\lambda$ . We use the normalized total variation norm

$$\|\mu - \rho\|_{\text{TV}} := \sup_{A \in \mathcal{B}(E)} |\mu(A) - \rho(A)|.$$

**Lemma 3.6** (Elementary Hellinger facts). *For probability measures  $\mu, \rho$ ,*

$$\mu \perp \rho \iff \|\mu - \rho\|_{\text{TV}} = 1 \iff A(\mu, \rho) = 0,$$

and

$$\|\mu - \rho\|_{\text{TV}} \leq \sqrt{1 - A(\mu, \rho)^2}.$$

Moreover, if  $\mathcal{P}$  ranges over finite measurable partitions of  $E$ , then

$$A(\mu, \rho) = \inf_{\mathcal{P}} \sum_{B \in \mathcal{P}} \sqrt{\mu(B)\rho(B)}.$$

*Proof.* Take  $\lambda = \mu + \rho$ , and write  $f = d\mu/d\lambda$ ,  $g = d\rho/d\lambda$ . The definition is independent of  $\lambda$  because changing the dominating measure multiplies both densities by the same Radon–Nikodym factor. If  $\mu \perp \rho$ , then  $fg = 0$   $\lambda$ -a.e., so  $A(\mu, \rho) = 0$ . Conversely, if  $A(\mu, \rho) = 0$ , then  $\sqrt{fg} = 0$   $\lambda$ -a.e.; the sets  $\{f > 0\}$  and  $\{g > 0\}$  separate  $\mu$  and  $\rho$ , so  $\mu \perp \rho$ . The equivalence with  $\|\mu - \rho\|_{\text{TV}} = 1$  is the standard characterization of singularity for probability measures.

Also,

$$\begin{aligned} \|\mu - \rho\|_{\text{TV}} &= \frac{1}{2} \int_E |f - g| d\lambda \\ &= \frac{1}{2} \int_E |\sqrt{f} - \sqrt{g}|(\sqrt{f} + \sqrt{g}) d\lambda \\ &\leq \frac{1}{2} \left( \int_E (\sqrt{f} - \sqrt{g})^2 d\lambda \right)^{1/2} \left( \int_E (\sqrt{f} + \sqrt{g})^2 d\lambda \right)^{1/2} \\ &= \sqrt{1 - A(\mu, \rho)^2}. \end{aligned}$$

For the partition formula, let  $\mathcal{P}$  be finite. On each atom  $B$ ,

$$\int_B \sqrt{\mathbb{E}_\lambda(f | \mathcal{P}) \mathbb{E}_\lambda(g | \mathcal{P})} d\lambda = \sqrt{\mu(B)\rho(B)}.$$

Conditional Cauchy–Schwarz gives

$$\mathbb{E}_\lambda(\sqrt{fg} | \mathcal{P}) \leq \sqrt{\mathbb{E}_\lambda(f | \mathcal{P}) \mathbb{E}_\lambda(g | \mathcal{P})},$$

so every partition sum is at least  $A(\mu, \rho)$ . Since  $E$  is Polish,  $\mathcal{B}(E)$  is countably generated. Choosing finite partitions increasing to a countable generating algebra, the bounded martingale convergence theorem (applied with  $\lambda = \mu + \rho$ , so  $0 \leq f, g \leq 1$ ) gives partition sums converging down to  $\int \sqrt{fg} d\lambda$ . Hence the infimum equals  $A(\mu, \rho)$ .  $\square$

**Lemma 3.7** (Mixture lower bound). *Let  $K$  be a Markov kernel on  $E$ . Suppose  $C \subset E$  is measurable and there is  $a > 0$  such that*

$$A(K(v, \cdot), K(w, \cdot)) \geq a \quad \forall v, w \in C.$$

Then, for all probability measures  $\alpha, \beta$ ,

$$A(\alpha K, \beta K) \geq a \alpha(C) \beta(C).$$

*Proof.* Let  $\mathcal{P}$  be a finite partition. For each atom  $B$ ,

$$\sqrt{(\alpha K)(B)(\beta K)(B)} \geq \int_C \int_C \sqrt{K(v, B)K(w, B)} \alpha(dv) \beta(dw),$$

by Cauchy–Schwarz applied to the functions  $\sqrt{K(v, B)}$  and  $\sqrt{K(w, B)}$ . Summing over  $B \in \mathcal{P}$  gives

$$\sum_{B \in \mathcal{P}} \sqrt{(\alpha K)(B)(\beta K)(B)} \geq \int_C \int_C \sum_{B \in \mathcal{P}} \sqrt{K(v, B)K(w, B)} \alpha(dv) \beta(dw).$$

The inner sum is at least  $A(K(v, \cdot), K(w, \cdot))$  by Lemma 3.6; hence the last display is at least  $a\alpha(C)\beta(C)$ . Taking the infimum over  $\mathcal{P}$  and using the partition formula proves the claim.  $\square$

**Lemma 3.8** (Lower semicontinuity of total variation). *If  $\mu_n \Rightarrow \mu$  and  $\rho_n \Rightarrow \rho$  weakly on a Polish space, then*

$$\|\mu - \rho\|_{\text{TV}} \leq \liminf_{n \rightarrow \infty} \|\mu_n - \rho_n\|_{\text{TV}}.$$

*Proof.* For  $\eta = \mu - \rho$ , regularity of the finite signed measure  $\eta$  implies

$$\|\eta\|_{\text{TV}} = \sup_{\substack{f \in C_b(E) \\ 0 \leq f \leq 1}} \left| \int_E f d\eta \right|.$$

Indeed, indicators of Borel sets are approximated from inside and outside by continuous functions using regularity of  $|\eta|$  and Urysohn’s lemma. For each such  $f$ ,

$$\int_E f d(\mu_n - \rho_n) \longrightarrow \int_E f d(\mu - \rho),$$

so

$$\left| \int_E f d(\mu - \rho) \right| \leq \liminf_n \|\mu_n - \rho_n\|_{\text{TV}}.$$

Taking the supremum over  $f$  proves the result.  $\square$

**Theorem 3.9** (Conditional uniqueness from compact-local Hellinger overlap). *Let  $E$  be Polish. Let  $(P_s)$  and  $(P_s^n)$  be Markov semigroups on  $E$ . Fix  $0 < \varepsilon < t$ . Assume:*

1. *for every  $x \in E$ ,*

$$P_t^n(x, \cdot) \Rightarrow P_t(x, \cdot), \quad P_{t-\varepsilon}^n(x, \cdot) \Rightarrow P_{t-\varepsilon}(x, \cdot);$$

2. *for every compact  $C \subset E$ , there is  $a(C) > 0$  such that*

$$A(P_\varepsilon^n(v, \cdot), P_\varepsilon^n(w, \cdot)) \geq a(C)$$

*for all  $n \geq 1$  and all  $v, w \in C$ .*

*Then  $P_t(x, \cdot) \not\equiv P_t(y, \cdot)$  for all  $x, y \in E$ . Consequently, if this holds for some  $t > 0$ , then  $(P_s)$  has at most one invariant probability measure.*

*Proof.* Fix  $x, y \in E$ , and set

$$\alpha_n := P_{t-\varepsilon}^n(x, \cdot), \quad \beta_n := P_{t-\varepsilon}^n(y, \cdot).$$

The two sequences are tight, because they are weakly convergent. Hence there is a compact  $C \subset E$  such that  $\alpha_n(C) \geq 1/2$  and  $\beta_n(C) \geq 1/2$  for all  $n$ . By hypothesis 2,

$$A(P_\varepsilon^n(v, \cdot), P_\varepsilon^n(w, \cdot)) \geq a(C), \quad v, w \in C.$$

Using the semigroup property and Lemma 3.7,

$$A(P_t^n(x, \cdot), P_t^n(y, \cdot)) = A(\alpha_n P_\varepsilon^n, \beta_n P_\varepsilon^n) \geq a(C) \alpha_n(C) \beta_n(C) \geq \frac{a(C)}{4}.$$

By Lemma 3.6,

$$\|P_t^n(x, \cdot) - P_t^n(y, \cdot)\|_{\text{TV}} \leq \sqrt{1 - \left(\frac{a(C)}{4}\right)^2} =: \rho < 1.$$

Weak convergence at time  $t$ , followed by Lemma 3.8, yields

$$\|P_t(x, \cdot) - P_t(y, \cdot)\|_{\text{TV}} \leq \rho < 1.$$

Thus the two limiting transition probabilities are not singular. The final claim follows from Theorem 3.4.  $\square$

### 3.4 Absolute continuity routes and the Dirichlet-form limitation

**Theorem 3.10** (Conditional uniqueness from compact-local uniform absolute continuity). *Let  $E = C_0([0, 1])$ , and let  $\nu$  be the Gibbs measure in (3.1). Fix  $t_0 > 0$ . Assume:*

1. *for every  $x \in E$  and  $f \in C_b(E)$ ,*

$$P_{t_0}^n f(x) \rightarrow P_{t_0} f(x);$$

2. *for every compact  $K \subset E$ , the family*

$$\{P_{t_0}^n(x, \cdot) : n \geq 1, x \in K\}$$

*is uniformly absolutely continuous with respect to  $\nu$ ;*

3. *every  $P_t$ -invariant probability measure that is absolutely continuous with respect to  $\nu$  equals  $\nu$ .*

*Then  $\nu$  is the unique invariant probability measure of  $(P_t)$ .*

*Proof.* Let  $\mu$  be invariant for  $(P_t)$ , and set  $\rho_n := \mu P_{t_0}^n$ . For  $f \in C_b(E)$ , dominated convergence and hypothesis 1 give

$$\int_E f d\rho_n = \int_E P_{t_0}^n f(x) \mu(dx) \longrightarrow \int_E P_{t_0} f(x) \mu(dx) = \int_E f d\mu,$$

so  $\rho_n \Rightarrow \mu$ .

Let  $B$  be Borel with  $\nu(B) = 0$ . Fix  $\varepsilon, \eta > 0$ . By tightness of  $\mu$ , choose compact  $K$  with  $\mu(K^c) < \eta$ . By uniform absolute continuity on  $K$ , choose  $\delta > 0$  such that

$$\nu(A) < \delta \implies \sup_{n \geq 1} \sup_{x \in K} P_{t_0}^n(x, A) < \varepsilon.$$

By outer regularity, choose open  $O \supset B$  with  $\nu(O) < \delta$ . Then for every  $n$ ,

$$\rho_n(O) = \int_E P_{t_0}^n(x, O) \mu(dx) \leq \varepsilon + \eta.$$

Since  $O$  is open and  $\rho_n \Rightarrow \mu$ , Portmanteau gives

$$\mu(O) \leq \liminf_n \rho_n(O) \leq \varepsilon + \eta.$$

Thus  $\mu(B) \leq \varepsilon + \eta$ . Letting  $\varepsilon, \eta \downarrow 0$  shows  $\mu \ll \nu$ . Hypothesis 3 gives  $\mu = \nu$ .  $\square$

*Remark 3.11* (A simple sufficient condition for uniform absolute continuity). Hypothesis 2 of Theorem 3.10 follows if, for every compact  $K \subset E$ , there are  $p > 1$  and  $C_K < \infty$  such that  $P_{t_0}^n(x, \cdot) = r_{n,x}\nu$  and

$$\sup_{n \geq 1} \sup_{x \in K} \|r_{n,x}\|_{L^p(\nu)} \leq C_K.$$

Indeed, with  $q = p/(p-1)$ , Holder's inequality gives  $P_{t_0}^n(x, A) \leq C_K \nu(A)^{1/q}$ .

**Proposition 3.12** (Pointwise resolvent absolute continuity). *Assume that  $\nu$  is invariant and that every invariant probability measure absolutely continuous with respect to  $\nu$  equals  $\nu$ . If, for some  $\lambda > 0$ ,*

$$R_\lambda(x, \cdot) := \int_0^\infty e^{-\lambda t} P_t(x, \cdot) dt \ll \nu \quad \forall x \in E,$$

*then  $\nu$  is the unique invariant probability measure.*

*Proof.* Let  $\mu$  be invariant and let  $B$  be Borel with  $\nu(B) = 0$ . Then  $R_\lambda(x, B) = 0$  for every  $x$ , and therefore

$$0 = \int_E R_\lambda(x, B) \mu(dx) = \int_0^\infty e^{-\lambda t} \mu P_t(B) dt = \int_0^\infty e^{-\lambda t} \mu(B) dt = \frac{1}{\lambda} \mu(B).$$

Thus  $\mu \ll \nu$ , and by the assumed uniqueness among  $\nu$ -absolutely continuous invariant measures,  $\mu = \nu$ .  $\square$

**Proposition 3.13** (What the Gibbs Dirichlet form proves). *Let  $(T_t)$  be a  $\nu$ -symmetric Markov semigroup on  $L^2(\nu)$ , and let  $\mathcal{E}$  be its closed Dirichlet form. Assume that the only zero-energy functions in the domain of  $\mathcal{E}$  are constants. Then every  $T_t$ -invariant probability measure  $\mu$  with  $\mu \ll \nu$  equals  $\nu$ .*

*For the Gibbs measure above, the zero-energy condition holds for the gradient form*

$$\mathcal{E}(F, G) = \frac{1}{2} \int_E \langle DF, DG \rangle_H d\nu$$

*provided the limiting semigroup is indeed the symmetric semigroup associated with this form.*

*Proof.* Let  $h = d\mu/d\nu$ . Invariance gives, for bounded  $f$ ,

$$\int_E T_t f h d\nu = \int_E f h d\nu.$$

By symmetry and the  $L^1$ -extension of  $T_t$ , this says  $T_t h = h$  in  $L^1(\nu)$ . For  $m \geq 1$ , let  $h_m = h \wedge m$ . Since  $r \mapsto r \wedge m$  is concave and  $T_t$  is Markov,

$$T_t h_m \leq (T_t h) \wedge m = h_m.$$

Integrating both sides against the invariant measure  $\nu$  gives equality, hence  $T_t h_m = h_m$ . Now  $h_m \in L^2(\nu)$ , and a fixed  $L^2$ -function has zero energy:

$$\mathcal{E}(h_m, h_m) = \lim_{t \downarrow 0} \frac{1}{t} \langle h_m - T_t h_m, h_m \rangle_{L^2(\nu)} = 0.$$

Thus  $h_m$  is constant. By monotone convergence,  $\int h_m d\nu \rightarrow \int h d\nu = 1$ , so  $h_m \rightarrow 1$  and  $h = 1$   $\nu$ -a.e.

For the displayed Gibbs gradient form, the density bounds (3.2) make  $\nu$  equivalent to  $\gamma$  with bounded density above and below. The Gaussian Poincare inequality for the Brownian-bridge measure,

$$\text{Var}_\gamma(F) \leq \frac{1}{\pi^2} \int_E \|DF\|_H^2 d\gamma$$

on cylinder functions, extends by closure. Hence zero  $\nu$ -energy implies zero  $\gamma$ -energy and therefore constancy.  $\square$

**Example 3.14** (Irreducibility in  $L^2(\nu)$  does not rule out singular invariant laws). Let  $E = [0, 1]$ , let  $\nu$  be Lebesgue measure, and define a Markov kernel

$$K(x, \cdot) = \begin{cases} \nu(\cdot), & x \neq 0, \\ \delta_0(\cdot), & x = 0. \end{cases}$$

Then  $\nu K = \nu$  because  $\nu(\{0\}) = 0$ , and  $\delta_0 K = \delta_0$ . On  $L^2(\nu)$ , however,  $Kf = \int f d\nu$   $\nu$ -a.e.; hence the  $L^2(\nu)$  operator is symmetric and has only constant fixed functions. The extra invariant measure  $\delta_0$  is invisible to the  $L^2(\nu)$  theory because it is supported on a  $\nu$ -null point.

### 3.5 What is missing from the stated assumptions

The preceding results identify exactly what would close the uniqueness proof. One needs a deterministic-start smoothing or overlap estimate, such as:

$$P_{t_0}(x, \cdot) \not\leq P_{t_0}(y, \cdot) \quad \forall x, y,$$

or the compact-local Hellinger estimate of Theorem 3.9, or compact-local uniform  $\nu$ -absolute continuity as in Theorem 3.10, or pointwise resolvent absolute continuity as in Proposition 3.12. These are not consequences of weak convergence alone.

**Example 3.15** (Weak convergence can destroy absolute continuity and overlap). On  $\mathbb{R}$ , each Gaussian law  $\mathcal{N}(0, 1/n)$  is absolutely continuous with respect to Lebesgue measure, but

$$\mathcal{N}(0, 1/n) \Rightarrow \delta_0,$$

and  $\delta_0$  is singular with respect to Lebesgue measure. Likewise,  $\mathcal{N}(0, 1/n)$  and  $\mathcal{N}(1, 1/n)$  have positive Hellinger affinity for each  $n$ , but their affinities tend to 0, and the weak limits  $\delta_0$  and  $\delta_1$  are singular. Thus weak convergence of kernels does not by itself preserve absolute continuity, uniform integrability, or a positive overlap lower bound.

*Remark 3.16* (Finite-dimensional domination as a possible missing theorem). A convenient way to prove the compact-local Hellinger input would be a uniform finite-dimensional heat-kernel domination theorem. Let  $P_t^{n,m}$  be the  $m$ -mode Galerkin approximation of the mollified equation and  $Q_t^m$  the  $m$ -dimensional Dirichlet Ornstein–Uhlenbeck kernel. A bound of the form

$$P_t^{n,m}(a, \cdot) \geq c Q_t^m(a, \cdot),$$

with  $c > 0$  independent of  $n, m, a$ , together with a passage from Galerkin kernels to the infinite-dimensional kernels, would yield overlap. For the Galerkin potential

$$\Psi_{n,m}(a) = \int_0^1 F_n \left( \sum_{k=1}^m a_k e_k(x) \right) dx,$$

one has the layer-cake representation, with  $e(x) = (e_1(x), \dots, e_m(x))$ ,

$$\Psi_{n,m}(a) = \int_0^1 \int_{\mathbb{R}} \mathbf{1}_{\{\langle e(x), a \rangle > s\}} b_n(s) ds \, dx.$$

This displays  $\Psi_{n,m}$  as an average of affine half-space membrane potentials of total mass 1. Proving the domination estimate is, however, an additional finite-dimensional heat-kernel theorem; it is not part of Assumption 3.1.

## Final conclusion

The natural Gibbs measure  $\nu$  is rigorously proved invariant for the limiting singular semigroup. The desired uniqueness statement follows from any of the deterministic-start smoothing or overlap estimates isolated above. Under only the assumptions explicitly stated in the original problem—namely well-posedness and weak convergence of the mollified equations—the argument does not rule out invariant measures singular with respect to  $\nu$ . Thus the candidate proof establishes invariance and conditional uniqueness, but an additional smoothing or overlap theorem is needed for an unconditional proof of (U).

## References

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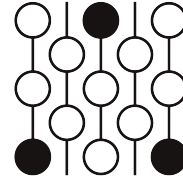


# Review of Submission 5B

First Proof Second Batch

Reviewer 3

June 4–8, 2026



## 1 Review

**Correctness.** The submission is not complete, and the last page admits that it does not solve the problem. The submission identifies two approaches to solving the problem - one by establishing Hellinger overlap and another proving that every invariant measure is absolutely continuous with respect to an invariant measure that can be constructed explicitly.

**Novelty.** The submission develops quite a bit of preliminary set-up for attacking the problem, but other than the last section there is not anything that is not found in graduate-level textbooks on SPDEs and ergodic theory.

The final section makes some interesting observations on possible ways to prove Hellinger overlap but these are far from worked out. To this degree, the submission contains a bit of novelty.

## 2 Recommendation

The submission makes little significant progress on solving the problem and while it contains some interesting ideas, there is nothing rigorous that is truly novel. Therefore the submission should be **rejected**.

## 3 Submission with inline comments

### Overview / Strategy

The original target is to decide whether the Markov semigroup  $P_t$  of the Dirichlet stochastic heat equation with distributional drift  $\delta_0$  has at most one invariant probability measure. The first part of the argument constructs the natural Gibbs measure

$$\nu(dv) = Z^{-1} \exp\left(2 \int_0^1 \mathbf{1}_{\{v(x) > 0\}} dx\right) \gamma(dv),$$

where  $\gamma$  is Brownian-bridge law, and proves that  $\nu$  is invariant by passing to the limit from the smooth mollified equations. This is carried out in Theorem 3.3.

Uniqueness requires more than this invariant-measure construction. The clean abstract mechanism is that pairwise non-singularity of one transition skeleton implies uniqueness; see Theorem 3.4. A compact-local Hellinger overlap estimate for the mollified last-step kernels would imply such non-singularity in the limit; see Theorem 3.9. A second route is to prove that every invariant measure is absolutely continuous with respect to  $\nu$ ; see Theorem 3.10 and Proposition 3.12. The stated ABLM-type assumptions provide existence, uniqueness in law, and weak convergence of the mollified equations, but they do not themselves provide any of these deterministic-start smoothing or overlap estimates. Thus the unconditional conclusion proved here is invariance of  $\nu$ , together with precise conditional criteria under which the desired uniqueness follows.

### 3.1 Standing assumptions, notation, and the target statement

Let

$$E := C_0([0, 1]) = \{v \in C([0, 1], \mathbb{R}) : v(0) = v(1) = 0\}, \quad H := L^2(0, 1),$$

with  $E$  equipped with the supremum norm. Let

$$e_k(x) := \sqrt{2} \sin(k\pi x), \quad \lambda_k := \pi^2 k^2, \quad k \geq 1,$$

be the Dirichlet eigenbasis of  $-\partial_{xx}$ . Let  $\gamma$  denote Brownian-bridge law on  $E$ , equivalently the centered Gaussian measure with covariance  $(-\partial_{xx})^{-1}$ :

$$\mathbb{E}_\gamma[(v, e_k)_H (v, e_\ell)_H] = \frac{\delta_{k\ell}}{\lambda_k}.$$

The SPDE from the original problem is

$$\begin{aligned} \frac{\partial u}{\partial t}(x, t) &= \frac{1}{2} \frac{\partial^2 u}{\partial x^2}(x, t) + \delta_0(u(x, t)) + \dot{W}(x, t), \quad t \geq 0, \\ u(x, 0) &= u_0(x) \in E, \\ u(0, t) &= u(1, t) = 0. \end{aligned} \tag{*}$$

**Assumption 3.1** (ABLM-type input on the bounded interval). For every  $u_0 \in E$  and  $T > 0$ , there is a unique-in-law adapted mild solution  $u \in C([0, T], E)$  to (\*), and the solutions define a Markov semigroup  $(P_t)_{t \geq 0}$  on  $E$ . For

$$b_n(r) := G_{1/n}(r) = \sqrt{\frac{n}{2\pi}} e^{-nr^2/2},$$

let  $P_t^n$  be the semigroup of the mollified equation obtained by replacing  $\delta_0$  with  $b_n$ . For every  $t > 0$ ,  $v \in E$ , and  $f \in C_b(E)$ ,

$$P_t^n f(v) \longrightarrow P_t f(v),$$

or equivalently  $P_t^n(v, \cdot) \Rightarrow P_t(v, \cdot)$ .

Define

$$F_n(r) := \int_{-\infty}^r b_n(s) ds, \quad \Phi_n(v) := \int_0^1 F_n(v(x)) dx, \quad \Phi(v) := \int_0^1 \mathbf{1}_{\{v(x)>0\}} dx.$$

Set

$$\nu_n(dv) := Z_n^{-1} e^{2\Phi_n(v)} \gamma(dv), \quad \nu(dv) := Z^{-1} e^{2\Phi(v)} \gamma(dv), \quad (3.1)$$

where  $Z_n$  and  $Z$  are the normalizing constants. Since  $0 \leq \Phi_n, \Phi \leq 1$ ,

$$1 \leq Z_n, Z \leq e^2, \quad e^{-2} \leq \frac{d\nu_n}{d\gamma}, \frac{d\nu}{d\gamma} \leq e^2. \quad (3.2)$$

For a Markov kernel  $K$  and a probability measure  $\mu$ , write  $\mu K(A) := \int_E K(x, A) \mu(dx)$ . A probability measure  $\mu$  is invariant for a semigroup  $(P_t)$  if  $\mu P_t = \mu$  for all  $t \geq 0$ .

**Theorem 3.2** (Original problem: target and established claim). *The target assertion from the original problem is*

$$(P_t) \text{ has at most one invariant probability measure on } E. \quad (\text{U})$$

*Under Assumption 3.1, the Gibbs measure  $\nu$  in (3.1) is invariant. Moreover, (U) follows if one adds any of the deterministic-start smoothing or overlap hypotheses in Theorem 3.9, Theorem 3.10, or Proposition 3.12 together with Proposition 3.13. In that case the unique invariant probability measure is  $\nu$ .*

*Proof.* The invariance statement is Theorem 3.3. The Hellinger overlap hypothesis of Theorem 3.9 implies pairwise non-singularity of one skeleton, and Theorem 3.4 then gives uniqueness. The compact-local uniform absolute-continuity hypothesis of Theorem 3.10 gives  $\mu \ll \nu$  for every invariant  $\mu$ , and its third hypothesis, supplied for instance by Proposition 3.13, then gives  $\mu = \nu$ . Finally, Proposition 3.12 is a resolvent version of the same absolute-continuity route. Since Theorem 3.3 supplies an invariant measure, any “at most one” conclusion identifies the invariant law with  $\nu$ .  $\square$

## 3.2 The Gibbs invariant measure

**Theorem 3.3** (Invariance of the Gibbs measure). *Under Assumption 3.1, the measure*

$$\nu(dv) = Z^{-1} \exp\left(2 \int_0^1 \mathbf{1}_{\{v(x)>0\}} dx\right) \gamma(dv)$$

*is invariant for  $(P_t)$ .*

*Proof.* We prove this by approximation.

*Step 1:  $\nu_n \rightarrow \nu$  in total variation.* For each  $r \neq 0$ ,  $F_n(r) \rightarrow \mathbf{1}_{\{r>0\}}$ . Let  $B \sim \gamma$ . For  $x \in (0, 1)$ ,  $B(x)$  is a nondegenerate centered Gaussian, hence  $\gamma(B(x) = 0) = 0$ . Therefore, by Fubini,

$$\mathbb{E}_\gamma \left| \{x \in [0, 1] : B(x) = 0\} \right| = \int_0^1 \gamma(B(x) = 0) dx = 0,$$

where the endpoints do not matter. Thus  $|\{x : B(x) = 0\}| = 0$  for  $\gamma$ -a.e.  $B$ . For such  $B$ ,

$$\Phi_n(B) = \int_0^1 F_n(B(x)) dx \longrightarrow \int_0^1 \mathbf{1}_{\{B(x) > 0\}} dx = \Phi(B)$$

by dominated convergence in  $x$ . Since  $0 \leq \Phi_n, \Phi \leq 1$ , dominated convergence in  $\gamma$  gives  $Z_n \rightarrow Z$  and

$$\int_E |Z_n^{-1} e^{2\Phi_n(v)} - Z^{-1} e^{2\Phi(v)}| \gamma(dv) \longrightarrow 0.$$

Hence  $\|\nu_n - \nu\|_{\text{TV}} \rightarrow 0$ .

*Step 2:  $\nu_n$  is invariant for  $P_t^n$ .* For fixed  $n$ , the drift  $b_n$  is bounded and Lipschitz. Also  $\Phi_n$  has  $H$ -gradient  $D\Phi_n(v) = b_n(v)$ , because for  $h \in H$ ,

$$D\Phi_n(v)[h] = \int_0^1 b_n(v(x))h(x) dx = \langle b_n(v), h \rangle_H.$$

On smooth cylinder functions the formal generator is

$$L_n F(v) = \frac{1}{2} \text{Tr } D^2 F(v) - \frac{1}{2} \langle -\partial_{xx} v, DF(v) \rangle_H + \langle D\Phi_n(v), DF(v) \rangle_H.$$

The Gaussian integration-by-parts formula for  $\gamma$ , together with  $d\nu_n = Z_n^{-1} e^{2\Phi_n} d\gamma$ , gives for smooth cylinder  $F, G$

$$\int_E G L_n F d\nu_n = -\frac{1}{2} \int_E \langle DG, DF \rangle_H d\nu_n.$$

Indeed, the derivative of  $e^{2\Phi_n}$  contributes  $2e^{2\Phi_n} D\Phi_n$ , which cancels exactly the added drift term  $\langle D\Phi_n, DF \rangle_H$ . Taking  $G \equiv 1$  gives

$$\int_E L_n F d\nu_n = 0.$$

For bounded Lipschitz gradient drifts, the standard martingale-problem or symmetric-Dirichlet-form construction of the stochastic heat equation identifies  $P_t^n$  with the Markov semigroup generated by the closure of  $L_n$ ; see, for example, [3, 4]. Infinitesimal invariance on the cylinder core therefore implies

$$\nu_n P_t^n = \nu_n, \quad t \geq 0.$$

*Step 3: pass to the singular limit.* Let  $f \in C_b(E)$  and  $t > 0$ . By Assumption 3.1,  $P_t^n f(v) \rightarrow P_t f(v)$  for every  $v \in E$ . Using  $d\nu_n/d\gamma \leq e^2$ , dominated convergence gives

$$\int_E |P_t^n f - P_t f| d\nu_n \leq e^2 \int_E |P_t^n f - P_t f| d\gamma \longrightarrow 0.$$

Also,  $\nu_n \rightarrow \nu$  in total variation and  $P_t f$  is bounded, so

$$\int_E P_t f d\nu_n \longrightarrow \int_E P_t f d\nu, \quad \int_E f d\nu_n \longrightarrow \int_E f d\nu.$$

Since  $\nu_n P_t^n = \nu_n$ ,

$$\int_E P_t f d\nu = \lim_{n \rightarrow \infty} \int_E P_t^n f d\nu_n = \lim_{n \rightarrow \infty} \int_E f d\nu_n = \int_E f d\nu.$$

Thus  $\nu P_t = \nu$  for all  $t > 0$ , and the case  $t = 0$  is trivial.  $\square$

### 3.3 Pairwise overlap and Hellinger criteria

**Theorem 3.4** (Pairwise non-singularity of one skeleton implies uniqueness). *Let  $E$  be Polish and  $K$  a Markov kernel on  $E$ . If*

$$K(x, \cdot) \not\perp K(y, \cdot) \quad \forall x, y \in E,$$

*then  $K$  has at most one invariant probability measure. Consequently, if a semigroup  $(P_t)$  has some  $t_0 > 0$  such that  $P_{t_0}(x, \cdot) \not\perp P_{t_0}(y, \cdot)$  for all  $x, y$ , then  $(P_t)$  has at most one invariant probability measure.*

*Proof.* Assume that  $\mu_1 \neq \mu_2$  are invariant for  $K$ . Let  $\sigma := \mu_1 - \mu_2$ , and write its Jordan decomposition  $\sigma = \sigma^+ - \sigma^-$ . Since  $\sigma(E) = 0$ ,  $m := \sigma^+(E) = \sigma^-(E) > 0$ .

For every finite signed measure  $\eta$ ,

$$|\eta K| \leq |\eta|K.$$

Indeed, for Borel  $A$ , taking the supremum over finite measurable partitions  $\{A_i\}$  of  $A$ ,

$$|\eta K|(A) \leq \sup_{\{A_i\}} \sum_i \int_E K(x, A_i) |\eta|(dx) = \int_E K(x, A) |\eta|(dx).$$

Since  $\sigma K = \sigma$ , we get  $|\sigma| \leq |\sigma|K$ . The two measures have equal total mass, so  $|\sigma|K = |\sigma|$ . Hence

$$\sigma^+ K = \frac{1}{2}(|\sigma|K + \sigma K) = \sigma^+, \quad \sigma^- K = \frac{1}{2}(|\sigma|K - \sigma K) = \sigma^-.$$

Thus  $\alpha := \sigma^+/m$  and  $\beta := \sigma^-/m$  are invariant probabilities, and  $\alpha \perp \beta$ . Choose a Borel set  $A$  with  $\alpha(A) = 1$  and  $\beta(A) = 0$ . Invariance gives

$$1 = \int_E K(x, A) \alpha(dx), \quad 0 = \int_E K(y, A) \beta(dy).$$

Therefore  $K(x, A) = 1$  for  $\alpha$ -a.e.  $x$  and  $K(y, A) = 0$  for  $\beta$ -a.e.  $y$ . Choosing such  $x, y$ ,  $K(x, \cdot)$  is concentrated on  $A$  while  $K(y, \cdot)$  is concentrated on  $A^c$ , so  $K(x, \cdot) \perp K(y, \cdot)$ , a contradiction. The semigroup assertion follows because every invariant measure for  $(P_t)$  is invariant for the skeleton  $P_{t_0}$ .  $\square$

**Definition 3.5** (Hellinger affinity). For probability measures  $\mu, \rho$  on  $E$ , define

$$A(\mu, \rho) := \int_E \sqrt{\frac{d\mu}{d\lambda} \frac{d\rho}{d\lambda}} d\lambda,$$

where  $\lambda$  is any finite measure dominating both  $\mu$  and  $\rho$ . The value is independent of  $\lambda$ . We use the normalized total variation norm

$$\|\mu - \rho\|_{\text{TV}} := \sup_{A \in \mathcal{B}(E)} |\mu(A) - \rho(A)|.$$

**Lemma 3.6** (Elementary Hellinger facts). *For probability measures  $\mu, \rho$ ,*

$$\mu \perp \rho \iff \|\mu - \rho\|_{\text{TV}} = 1 \iff A(\mu, \rho) = 0,$$

and

$$\|\mu - \rho\|_{\text{TV}} \leq \sqrt{1 - A(\mu, \rho)^2}.$$

Moreover, if  $\mathcal{P}$  ranges over finite measurable partitions of  $E$ , then

$$A(\mu, \rho) = \inf_{\mathcal{P}} \sum_{B \in \mathcal{P}} \sqrt{\mu(B)\rho(B)}.$$

*Proof.* Take  $\lambda = \mu + \rho$ , and write  $f = d\mu/d\lambda$ ,  $g = d\rho/d\lambda$ . The definition is independent of  $\lambda$  because changing the dominating measure multiplies both densities by the same Radon–Nikodym factor. If  $\mu \perp \rho$ , then  $fg = 0$   $\lambda$ -a.e., so  $A(\mu, \rho) = 0$ . Conversely, if  $A(\mu, \rho) = 0$ , then  $\sqrt{fg} = 0$   $\lambda$ -a.e.; the sets  $\{f > 0\}$  and  $\{g > 0\}$  separate  $\mu$  and  $\rho$ , so  $\mu \perp \rho$ . The equivalence with  $\|\mu - \rho\|_{\text{TV}} = 1$  is the standard characterization of singularity for probability measures.

Also,

$$\begin{aligned} \|\mu - \rho\|_{\text{TV}} &= \frac{1}{2} \int_E |f - g| d\lambda \\ &= \frac{1}{2} \int_E |\sqrt{f} - \sqrt{g}|(\sqrt{f} + \sqrt{g}) d\lambda \\ &\leq \frac{1}{2} \left( \int_E (\sqrt{f} - \sqrt{g})^2 d\lambda \right)^{1/2} \left( \int_E (\sqrt{f} + \sqrt{g})^2 d\lambda \right)^{1/2} \\ &= \sqrt{1 - A(\mu, \rho)^2}. \end{aligned}$$

For the partition formula, let  $\mathcal{P}$  be finite. On each atom  $B$ ,

$$\int_B \sqrt{\mathbb{E}_\lambda(f | \mathcal{P}) \mathbb{E}_\lambda(g | \mathcal{P})} d\lambda = \sqrt{\mu(B)\rho(B)}.$$

Conditional Cauchy–Schwarz gives

$$\mathbb{E}_\lambda(\sqrt{fg} | \mathcal{P}) \leq \sqrt{\mathbb{E}_\lambda(f | \mathcal{P}) \mathbb{E}_\lambda(g | \mathcal{P})},$$

so every partition sum is at least  $A(\mu, \rho)$ . Since  $E$  is Polish,  $\mathcal{B}(E)$  is countably generated. Choosing finite partitions increasing to a countable generating algebra, the bounded martingale convergence theorem (applied with  $\lambda = \mu + \rho$ , so  $0 \leq f, g \leq 1$ ) gives partition sums converging down to  $\int \sqrt{fg} d\lambda$ . Hence the infimum equals  $A(\mu, \rho)$ .  $\square$

**Lemma 3.7** (Mixture lower bound). *Let  $K$  be a Markov kernel on  $E$ . Suppose  $C \subset E$  is measurable and there is  $a > 0$  such that*

$$A(K(v, \cdot), K(w, \cdot)) \geq a \quad \forall v, w \in C.$$

Then, for all probability measures  $\alpha, \beta$ ,

$$A(\alpha K, \beta K) \geq a \alpha(C) \beta(C).$$

*Proof.* Let  $\mathcal{P}$  be a finite partition. For each atom  $B$ ,

$$\sqrt{(\alpha K)(B)(\beta K)(B)} \geq \int_C \int_C \sqrt{K(v, B)K(w, B)} \alpha(dv) \beta(dw),$$

by Cauchy–Schwarz applied to the functions  $\sqrt{K(v, B)}$  and  $\sqrt{K(w, B)}$ . Summing over  $B \in \mathcal{P}$  gives

$$\sum_{B \in \mathcal{P}} \sqrt{(\alpha K)(B)(\beta K)(B)} \geq \int_C \int_C \sum_{B \in \mathcal{P}} \sqrt{K(v, B)K(w, B)} \alpha(dv) \beta(dw).$$

The inner sum is at least  $A(K(v, \cdot), K(w, \cdot))$  by Lemma 3.6; hence the last display is at least  $a\alpha(C)\beta(C)$ . Taking the infimum over  $\mathcal{P}$  and using the partition formula proves the claim.  $\square$

**Lemma 3.8** (Lower semicontinuity of total variation). *If  $\mu_n \Rightarrow \mu$  and  $\rho_n \Rightarrow \rho$  weakly on a Polish space, then*

$$\|\mu - \rho\|_{\text{TV}} \leq \liminf_{n \rightarrow \infty} \|\mu_n - \rho_n\|_{\text{TV}}.$$

*Proof.* For  $\eta = \mu - \rho$ , regularity of the finite signed measure  $\eta$  implies

$$\|\eta\|_{\text{TV}} = \sup_{\substack{f \in C_b(E) \\ 0 \leq f \leq 1}} \left| \int_E f d\eta \right|.$$

Indeed, indicators of Borel sets are approximated from inside and outside by continuous functions using regularity of  $|\eta|$  and Urysohn’s lemma. For each such  $f$ ,

$$\int_E f d(\mu_n - \rho_n) \longrightarrow \int_E f d(\mu - \rho),$$

so

$$\left| \int_E f d(\mu - \rho) \right| \leq \liminf_n \|\mu_n - \rho_n\|_{\text{TV}}.$$

Taking the supremum over  $f$  proves the result.  $\square$

**Theorem 3.9** (Conditional uniqueness from compact-local Hellinger overlap). *Let  $E$  be Polish. Let  $(P_s)$  and  $(P_s^n)$  be Markov semigroups on  $E$ . Fix  $0 < \varepsilon < t$ . Assume:*

1. *for every  $x \in E$ ,*

$$P_t^n(x, \cdot) \Rightarrow P_t(x, \cdot), \quad P_{t-\varepsilon}^n(x, \cdot) \Rightarrow P_{t-\varepsilon}(x, \cdot);$$

2. *for every compact  $C \subset E$ , there is  $a(C) > 0$  such that*

$$A(P_\varepsilon^n(v, \cdot), P_\varepsilon^n(w, \cdot)) \geq a(C)$$

*for all  $n \geq 1$  and all  $v, w \in C$ .*

*Then  $P_t(x, \cdot) \not\equiv P_t(y, \cdot)$  for all  $x, y \in E$ . Consequently, if this holds for some  $t > 0$ , then  $(P_s)$  has at most one invariant probability measure.*

*Proof.* Fix  $x, y \in E$ , and set

$$\alpha_n := P_{t-\varepsilon}^n(x, \cdot), \quad \beta_n := P_{t-\varepsilon}^n(y, \cdot).$$

The two sequences are tight, because they are weakly convergent. Hence there is a compact  $C \subset E$  such that  $\alpha_n(C) \geq 1/2$  and  $\beta_n(C) \geq 1/2$  for all  $n$ . By hypothesis 2,

$$A(P_\varepsilon^n(v, \cdot), P_\varepsilon^n(w, \cdot)) \geq a(C), \quad v, w \in C.$$

Using the semigroup property and Lemma 3.7,

$$A(P_t^n(x, \cdot), P_t^n(y, \cdot)) = A(\alpha_n P_\varepsilon^n, \beta_n P_\varepsilon^n) \geq a(C) \alpha_n(C) \beta_n(C) \geq \frac{a(C)}{4}.$$

By Lemma 3.6,

$$\|P_t^n(x, \cdot) - P_t^n(y, \cdot)\|_{\text{TV}} \leq \sqrt{1 - \left(\frac{a(C)}{4}\right)^2} =: \rho < 1.$$

Weak convergence at time  $t$ , followed by Lemma 3.8, yields

$$\|P_t(x, \cdot) - P_t(y, \cdot)\|_{\text{TV}} \leq \rho < 1.$$

Thus the two limiting transition probabilities are not singular. The final claim follows from Theorem 3.4.  $\square$

### 3.4 Absolute continuity routes and the Dirichlet-form limitation

**Theorem 3.10** (Conditional uniqueness from compact-local uniform absolute continuity). *Let  $E = C_0([0, 1])$ , and let  $\nu$  be the Gibbs measure in (3.1). Fix  $t_0 > 0$ . Assume:*

1. *for every  $x \in E$  and  $f \in C_b(E)$ ,*

$$P_{t_0}^n f(x) \rightarrow P_{t_0} f(x);$$

2. *for every compact  $K \subset E$ , the family*

$$\{P_{t_0}^n(x, \cdot) : n \geq 1, x \in K\}$$

*is uniformly absolutely continuous with respect to  $\nu$ ;*

3. *every  $P_t$ -invariant probability measure that is absolutely continuous with respect to  $\nu$  equals  $\nu$ .*

*Then  $\nu$  is the unique invariant probability measure of  $(P_t)$ .*



*Proof.* Let  $\mu$  be invariant for  $(P_t)$ , and set  $\rho_n := \mu P_{t_0}^n$ . For  $f \in C_b(E)$ , dominated convergence and hypothesis 1 give

$$\int_E f d\rho_n = \int_E P_{t_0}^n f(x) \mu(dx) \longrightarrow \int_E P_{t_0} f(x) \mu(dx) = \int_E f d\mu,$$

so  $\rho_n \Rightarrow \mu$ .

Let  $B$  be Borel with  $\nu(B) = 0$ . Fix  $\varepsilon, \eta > 0$ . By tightness of  $\mu$ , choose compact  $K$  with  $\mu(K^c) < \eta$ . By uniform absolute continuity on  $K$ , choose  $\delta > 0$  such that

$$\nu(A) < \delta \implies \sup_{n \geq 1} \sup_{x \in K} P_{t_0}^n(x, A) < \varepsilon.$$

By outer regularity, choose open  $O \supset B$  with  $\nu(O) < \delta$ . Then for every  $n$ ,

$$\rho_n(O) = \int_E P_{t_0}^n(x, O) \mu(dx) \leq \varepsilon + \eta.$$

Since  $O$  is open and  $\rho_n \Rightarrow \mu$ , Portmanteau gives

$$\mu(O) \leq \liminf_n \rho_n(O) \leq \varepsilon + \eta.$$

Thus  $\mu(B) \leq \varepsilon + \eta$ . Letting  $\varepsilon, \eta \downarrow 0$  shows  $\mu \ll \nu$ . Hypothesis 3 gives  $\mu = \nu$ .  $\square$

*Remark 3.11* (A simple sufficient condition for uniform absolute continuity). Hypothesis 2 of Theorem 3.10 follows if, for every compact  $K \subset E$ , there are  $p > 1$  and  $C_K < \infty$  such that  $P_{t_0}^n(x, \cdot) = r_{n,x}\nu$  and

$$\sup_{n \geq 1} \sup_{x \in K} \|r_{n,x}\|_{L^p(\nu)} \leq C_K.$$

Indeed, with  $q = p/(p-1)$ , Holder's inequality gives  $P_{t_0}^n(x, A) \leq C_K \nu(A)^{1/q}$ .

**Proposition 3.12** (Pointwise resolvent absolute continuity). *Assume that  $\nu$  is invariant and that every invariant probability measure absolutely continuous with respect to  $\nu$  equals  $\nu$ . If, for some  $\lambda > 0$ ,*

$$R_\lambda(x, \cdot) := \int_0^\infty e^{-\lambda t} P_t(x, \cdot) dt \ll \nu \quad \forall x \in E,$$

*then  $\nu$  is the unique invariant probability measure.*

*Proof.* Let  $\mu$  be invariant and let  $B$  be Borel with  $\nu(B) = 0$ . Then  $R_\lambda(x, B) = 0$  for every  $x$ , and therefore

$$0 = \int_E R_\lambda(x, B) \mu(dx) = \int_0^\infty e^{-\lambda t} \mu P_t(B) dt = \int_0^\infty e^{-\lambda t} \mu(B) dt = \frac{1}{\lambda} \mu(B).$$

Thus  $\mu \ll \nu$ , and by the assumed uniqueness among  $\nu$ -absolutely continuous invariant measures,  $\mu = \nu$ .  $\square$

**Proposition 3.13** (What the Gibbs Dirichlet form proves). *Let  $(T_t)$  be a  $\nu$ -symmetric Markov semigroup on  $L^2(\nu)$ , and let  $\mathcal{E}$  be its closed Dirichlet form. Assume that the only zero-energy functions in the domain of  $\mathcal{E}$  are constants. Then every  $T_t$ -invariant probability measure  $\mu$  with  $\mu \ll \nu$  equals  $\nu$ .*

*For the Gibbs measure above, the zero-energy condition holds for the gradient form*

$$\mathcal{E}(F, G) = \frac{1}{2} \int_E \langle DF, DG \rangle_H d\nu$$

*provided the limiting semigroup is indeed the symmetric semigroup associated with this form.*

*Proof.* Let  $h = d\mu/d\nu$ . Invariance gives, for bounded  $f$ ,

$$\int_E T_t f h d\nu = \int_E f h d\nu.$$

By symmetry and the  $L^1$ -extension of  $T_t$ , this says  $T_t h = h$  in  $L^1(\nu)$ . For  $m \geq 1$ , let  $h_m = h \wedge m$ . Since  $r \mapsto r \wedge m$  is concave and  $T_t$  is Markov,

$$T_t h_m \leq (T_t h) \wedge m = h_m.$$

Integrating both sides against the invariant measure  $\nu$  gives equality, hence  $T_t h_m = h_m$ . Now  $h_m \in L^2(\nu)$ , and a fixed  $L^2$ -function has zero energy:

$$\mathcal{E}(h_m, h_m) = \lim_{t \downarrow 0} \frac{1}{t} \langle h_m - T_t h_m, h_m \rangle_{L^2(\nu)} = 0.$$

Thus  $h_m$  is constant. By monotone convergence,  $\int h_m d\nu \rightarrow \int h d\nu = 1$ , so  $h_m \rightarrow 1$  and  $h = 1$   $\nu$ -a.e.

For the displayed Gibbs gradient form, the density bounds (3.2) make  $\nu$  equivalent to  $\gamma$  with bounded density above and below. The Gaussian Poincare inequality for the Brownian-bridge measure,

$$\text{Var}_\gamma(F) \leq \frac{1}{\pi^2} \int_E \|DF\|_H^2 d\gamma$$

on cylinder functions, extends by closure. Hence zero  $\nu$ -energy implies zero  $\gamma$ -energy and therefore constancy.  $\square$

**Example 3.14** (Irreducibility in  $L^2(\nu)$  does not rule out singular invariant laws). Let  $E = [0, 1]$ , let  $\nu$  be Lebesgue measure, and define a Markov kernel

$$K(x, \cdot) = \begin{cases} \nu(\cdot), & x \neq 0, \\ \delta_0(\cdot), & x = 0. \end{cases}$$

Then  $\nu K = \nu$  because  $\nu(\{0\}) = 0$ , and  $\delta_0 K = \delta_0$ . On  $L^2(\nu)$ , however,  $Kf = \int f d\nu$   $\nu$ -a.e.; hence the  $L^2(\nu)$  operator is symmetric and has only constant fixed functions. The extra invariant measure  $\delta_0$  is invisible to the  $L^2(\nu)$  theory because it is supported on a  $\nu$ -null point.

### 3.5 What is missing from the stated assumptions

The preceding results identify exactly what would close the uniqueness proof. One needs a deterministic-start smoothing or overlap estimate, such as:

$$P_{t_0}(x, \cdot) \not\leq P_{t_0}(y, \cdot) \quad \forall x, y,$$

or the compact-local Hellinger estimate of Theorem 3.9, or compact-local uniform  $\nu$ -absolute continuity as in Theorem 3.10, or pointwise resolvent absolute continuity as in Proposition 3.12. These are not consequences of weak convergence alone.

**Example 3.15** (Weak convergence can destroy absolute continuity and overlap). On  $\mathbb{R}$ , each Gaussian law  $\mathcal{N}(0, 1/n)$  is absolutely continuous with respect to Lebesgue measure, but

$$\mathcal{N}(0, 1/n) \Rightarrow \delta_0,$$

and  $\delta_0$  is singular with respect to Lebesgue measure. Likewise,  $\mathcal{N}(0, 1/n)$  and  $\mathcal{N}(1, 1/n)$  have positive Hellinger affinity for each  $n$ , but their affinities tend to 0, and the weak limits  $\delta_0$  and  $\delta_1$  are singular. Thus weak convergence of kernels does not by itself preserve absolute continuity, uniform integrability, or a positive overlap lower bound.

*Remark 3.16* (Finite-dimensional domination as a possible missing theorem). A convenient way to prove the compact-local Hellinger input would be a uniform finite-dimensional heat-kernel domination theorem. Let  $P_t^{n,m}$  be the  $m$ -mode Galerkin approximation of the mollified equation and  $Q_t^m$  the  $m$ -dimensional Dirichlet Ornstein–Uhlenbeck kernel. A bound of the form

$$P_t^{n,m}(a, \cdot) \geq c Q_t^m(a, \cdot),$$

with  $c > 0$  independent of  $n, m, a$ , together with a passage from Galerkin kernels to the infinite-dimensional kernels, would yield overlap. For the Galerkin potential

$$\Psi_{n,m}(a) = \int_0^1 F_n \left( \sum_{k=1}^m a_k e_k(x) \right) dx,$$

one has the layer-cake representation, with  $e(x) = (e_1(x), \dots, e_m(x))$ ,

$$\Psi_{n,m}(a) = \int_0^1 \int_{\mathbb{R}} \mathbf{1}_{\{\langle e(x), a \rangle > s\}} b_n(s) ds \, dx.$$

This displays  $\Psi_{n,m}$  as an average of affine half-space membrane potentials of total mass 1. Proving the domination estimate is, however, an additional finite-dimensional heat-kernel theorem; it is not part of Assumption 3.1.

## Final conclusion

The natural Gibbs measure  $\nu$  is rigorously proved invariant for the limiting singular semigroup. The desired uniqueness statement follows from any of the deterministic-start smoothing or overlap estimates isolated above. Under only the assumptions explicitly stated in the original problem—namely well-posedness and weak convergence of the mollified equations—the argument does not rule out invariant measures singular with respect to  $\nu$ . Thus the candidate proof establishes invariance and conditional uniqueness, but an additional smoothing or overlap theorem is needed for an unconditional proof of (U).

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